



Greenhouse gas emissions from small-scale fly larvae composting with *Hermetia illucens*

E. Ermolaev*, C. Lalander, B. Vinnerås

Swedish University of Agricultural Sciences (SLU), Department of Energy and Technology, Box 7032, 75007 Uppsala, Sweden

ARTICLE INFO

Article history:

Received 8 February 2019

Revised 3 July 2019

Accepted 5 July 2019

Available online 15 July 2019

Keywords:

Eco-innovation

BSF

Black soldier fly

Food waste

Pre-treatment

Waste management

ABSTRACT

Fly larvae composting is an emerging waste treatment alternative with great potential to increase revenue from food waste management. For wider implementation, fly larvae composting has to be evaluated in comparison with conventional systems, based on direct greenhouse gas (GHG) emission data for the treatment process, which are currently limited. This study evaluated direct emissions of CO₂, CH₄, N₂O and NH₃ from composting of food waste using black soldier fly (BSF) larvae (*Hermetia illucens*). Use of BSF larvae-associated bacteria in 7-day pre-treatment and seeding at larvae treatment start were evaluated and compared to larvae treatment without bacteria addition. The treatments were performed in a set of 14-day laboratory-scale experiments. Mean substrate reduction was 49 ± 8% and bioconversion ratio was 24 ± 8% (both dry matter basis). Direct GHG emissions from the fly larvae treatment process were generally very small, with emissions of CH₄ and N₂O equivalent to 0.38 kg CO₂-equivalents per ton food waste treated assuming global warming potential over 100 years, while mean total CO₂ emissions were 96 g CO₂ per kg food waste treated. Additional emissions could be expected to occur in the pre-treatment process, which did not provide any significant improvement in bioconversion ratio or reduction in total GHG emissions during treatment. Similarly, use of BSF larvae-associated bacteria did not significantly improve process efficiency. No NH₃ emissions were detected, as reflected in total N mass balance over the treatment cycle. The results show that total direct GHG emissions from food waste treatment by fly larvae composting are lower than those from conventional food waste treatment, and that pre-treatment and seeding of food waste with BSF larvae-associated bacteria do not further reduce total GHG emissions.

© 2019 Elsevier Ltd. All rights reserved.

1. Introduction

Fly larvae composting is a waste treatment method that has great potential to increase revenues from food waste management (Lalander et al., 2018). Fly larvae composting is an aerobic process (as is conventional composting) in which most of the energy available in the substrate is transformed into larval biomass (Lalander et al., 2015). This transformation is expected to result in lower CO₂ emissions, as the part of carbon accumulated in larval biomass is not lost (Perednia, 2017). When a life cycle assessment (LCA) approach is used for evaluating the environmental impacts of fly larvae composting, data on direct greenhouse gas (GHG) emissions are needed, as the contribution of the treatment step must be included when estimating the global warming potential (GWP) of the whole system (Salomone et al., 2017). The larval biomass generated in fly larvae composting systems is a valuable additional

product to the fertiliser which is typically generated in biological treatment of food waste. Implementation of fly larvae composting allows implementation of a system where food waste is viewed as a resource and generates high value products unlike in conventional systems where at best a partial recovery of energy and plant nutrients is possible. The larval biomass resulting from fly larvae composting can be used as feed for animals, further reducing the overall system emissions since other sources of feed (e.g. soybean) can be replaced (Allegretti et al., 2018).

Other potential emissions from the degradation process, such as methane (CH₄) and nitrous oxide (N₂O), may be a concern, as they are in conventional food waste treatment methods such as composting and anaerobic digestion (Møller et al., 2009; Pipatti et al., 2006). Emissions of CH₄ and N₂O are relevant to investigate, as they have GWP of 34 and 289, respectively, over 100 years (IPCC, 2013). Studies focusing on LCA of waste management systems and investigating fly larva composting as one of the alternatives highlighted knowledge gaps in inventory data on emissions from fly larvae composting as a food waste treatment (Allegretti et al.,

* Corresponding author.

E-mail address: evgheni.ermolaev@slu.se (E. Ermolaev).

2018; Mertenat et al., 2019; Salomone et al., 2017). These LCA studies use emission data for other insect species to overcome the lack of data. A recent LCA study by Mertenat et al. (2019) evaluated direct emissions from BSF treatment of food waste and found that they were lower than in conventional composting. However, further evaluation based on homogeneous feedstock and carbon and nitrogen balance is necessary to evaluate the effect of substrate and process parameters on the emissions. To our knowledge, direct CH₄ or N₂O emissions from fly larvae composting of food waste have not been determined to date under controlled laboratory conditions. Emissions of CH₄ typically occur due to methanogenic microbe activity under anaerobic conditions and can be expected to be low in fly larvae composting, due to larval movement mechanically turning the substrate and making it more aerobic (Cickova et al., 2015). Emissions of N₂O occur from processes such as nitrification and denitrification in aerobically degraded organic material and depend on both substrate properties and microbial activity (Ermolaev et al., 2015). Overall low N₂O emissions have been reported for *Neomyia cornicina* fly larvae treating cattle dung, an effect attributed to inhibition of microbial degradation activity and nitrogen assimilation within larval biomass (Iwasa et al., 2015). Fly larvae of *Hermetia illucens* have been reported to modify the microbiome of their surrounding substrate (Jeon et al., 2011; Varotto Boccazzi et al., 2017). However, the effects of this on emissions of N₂O have not been investigated.

Another relevant gas associated with aerobic degradation of organic material is ammonia (NH₃), as it contributes to acidification and eutrophication (IPCC, 2014). This gas can represent a major pathway of nitrogen losses in conventional composting (Jarvis et al., 2009). Ammonia has been reported to cause odours and even to create a hazardous environment for workers in fly larvae composting and can represent a major pathway for nitrogen losses from the compost produced (Cickova et al., 2015; Cickova et al., 2012; Lalander et al., 2015). The pathway for NH₃ emissions can be expected to be similar to that in conventional composting. Several factors are decisive for NH₃ emissions during composting; when aerobic degradation occurs at a high rate, organic nitrogen mineralisation produces ammonium (NH₄⁺) as the primary product (Paredes et al., 1996). Ammonium is converted to gaseous form (NH₃) when the substrate pH shifts towards alkaline and the rate of NH₃ emission increases with aeration and temperature (Rynk and Richard, 2001).

High efficiency in fly larva composting on optimal substrates makes this method attractive for use as a major food waste treatment method (Lalander et al., 2019). However, further process improvement is needed, as organic degradation in the process is not complete (Lalander et al., 2018). BSF larvae-associated bacteria have been suggested as a means to improve the efficiency in the fly larvae composting process (Yu et al., 2011). Pre-treatment of the substrate with these bacteria could potentially improve nutrient availability to the larvae, as parts of food waste are otherwise slow to degrade (Yu et al., 2011; Zheng et al., 2013). Alternatively, seeding the substrate with BSF-associated bacteria at the start of larvae treatment could accelerate the process of larval microbiome adjustment, while also stimulating degradation to some degree (Xiao et al., 2018; Yu et al., 2011).

The aim of this study was to evaluate GHG and NH₃ emissions from fly larvae composting treatment and the effects of pre-treatment and seeding of substrate with BSF larvae-associated bacteria on the efficiency and rate of the fly larvae composting process and the amount of emissions produced. The results of this study are key in performing further life cycle assessment (LCA) evaluation of fly larvae composting as a waste management alternative and for comparing its performance to other waste treatment alternatives, currently there is lack of such data.

2. Material and methods

2.1. Materials

In all treatments performed in this study, food waste was used as the substrate for the larvae. This food waste was sourced from a local canteen at SLU (Uppsala, Sweden) and represented a mixture of food preparation waste, food scraps and unsold prepared food. The food waste had a dry matter (DM) content of $24.9 \pm 2.9\%$ of wet weight (WW), volatile solids content of $92.3 \pm 0.5\%$ of DM, pH of 4.42 ± 0.63 and density of $1.09 \pm 0.06 \text{ g mL}^{-1}$ on average ($n = 3$). The collected substrate was homogenised, divided into feeding portions and stored at -20°C until use, prior to which it was thawed and allowed to reach room temperature for up to 12 h. In all treatments, 5-d-old, $>0.2 \text{ cm}$ long BSF larvae obtained from a colony continuously running since 2015 at SLU (Uppsala, Sweden) were used.

2.2. Study layout

All treatments were performed in 11 cm deep small boxes with dimensions of $21 \times 17 \text{ cm}$ giving a treatment surface of 357 cm^2 to which 700 larvae were added, giving a larval density of 2 larvae cm^{-2} . The larvae density and feeding rate of $30.0 \pm 0.3 \text{ mg dry matter larva}^{-1} \text{ d}^{-1}$ were selected to ensure sufficient nutrition and avoid overcrowding in accordance to previously established recommendations (Diener et al., 2011). The boxes were incubated at 28°C during the 14-day treatment period to obtain larvae in the fifth instar. The larvae were added by weight based on individual larvae weight determined on a smaller sample. Three treatments were compared: Treatment *L* was standard fly larvae composting, where only fly larvae were added to the food waste; in treatment *BL*, fly larvae and a spore-forming bacteria consortium isolated from the gut of BSF larvae were added at the start of the composting; in treatment *PBL*, the food waste was pre-treated in open boxes at 28°C for 7 days with the same bacteria consortium as in *BL*, after which fly larvae were added (Fig. 1). Spore-forming bacteria were chosen due to simplicity of their implementation as inoculum in future applications but also to avoid possibility of recycling pathogens. At the end of the experiments, the larvae were picked out, weighed and enumerated. Based on the final number of larvae, their survival was calculated in relation to the initial number. The treatment residues were weighed and sampled for analysis of dry matter content (DM), total volatile solids (VS) and total nitrogen (N). All treatments were performed in triplicate.

2.3. Bacterial isolation

Twelve BSF larvae (BSFL) reaching instar 5 (reared in food waste) were picked out and washed in deionised water and then in disinfection solution (70% EtOH). The disinfected larvae were manually chopped to pieces and immersed in 10 mL (4 BSFL/10 mL) nutrient broth (1 g/L Lab-Lemco powder, 2 g/L yeast extract, 5 g/L peptone and 5 g/L NaCl, National Veterinary Institute, Uppsala, Sweden) at 70°C for 10 min. After cooling to room temperature (RT), 1 mL of broth from each tube was spread on taurocholate gelatine agar (TGA) plates ($\varnothing 14 \text{ cm}$; Miclev, Sweden) and the plates were incubated at 37°C for 24 h. Five visually different colonies were clean-spread on TGA plates and incubated at 37°C for 24 h. Single individual colonies were taken and heat-treated once more in the nutrient broth (as described above). For each colony, upon reaching RT, $10 \mu\text{L}$ of broth were spread on a TGA plate and the plates were incubated at 37°C for 24 h. Once again, single colonies were collected from each plate and clean-spread on TGA plates that were incubated at 37°C for 24 h. The

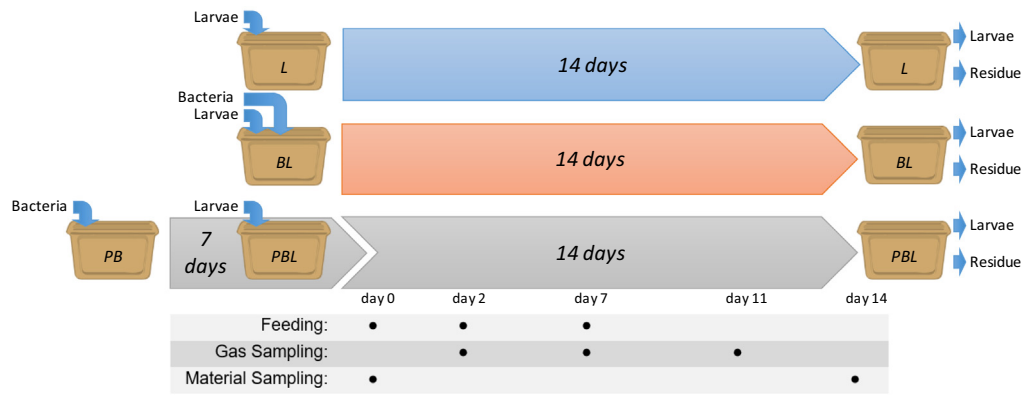


Fig. 1. Diagram of the study layout, showing the feeding and sampling schedule during the three treatments studied, in which food waste was treated with black soldier fly (BSF) larvae (treatment L), treated with BSF-associated bacteria and larvae (treatment BL) and pre-treated with BSF-associated bacteria and treated with larvae (treatment PBL).

clean-spread colonies were kept at 4 °C until use, upon which one colony from each clean-spread plate was added to 10 mL bacterial nutrient broth (National Veterinary Institute, Sweden) and incubated at 37 °C for 24 h. In total, five visually different bacterial colonies were used. No bacterial typing was performed for the isolated bacteria. At the start of treatment BL, 1% (w/w) of bacterial inoculate broth prepared from each colony was added to the food waste together with the BSF larvae. The concentration of 1% broth to food waste was to ensure a quick establishment of the microbial community being inoculated and to decrease the risk of the inoculum being outcompeted by naturally occurring bacteria. In treatment PBL, the same type and concentration of bacterial inoculate was added to food waste and incubated at 28 °C for 7 days prior to larvae addition (Fig. 1).

2.4. Material sampling

Substrate samples were collected from the initial food waste mixture and from residues left at the end of the treatments (Fig. 1). These were analysed gravimetrically for DM content by drying at 80 °C until stable weight and for VS content by incineration at 250 °C for 2 h, followed by 550 °C for 4 h. Homogenised food waste density was estimated by filling graduated cylinders and measuring the volume and weight of the substrate. For total nitrogen (total N) determination in food waste, 0.5 g of material was boiled in concentrated sulphuric acid (H₂SO₄), following the procedure described in Lalander et al. (2015). The samples were neutralised using 10 M sodium hydroxide (NaOH), diluted 1:50 in deionised water and digested using Spectroquant® Crack-Set 20 (1.14963.0001). The nitrate concentration in the digested solution was then determined at 340 nm using the Spectroquant® nitrate test (1.09713.0002). Larvae total N was analysed externally (Agri-Lab AB, Sweden) in dried ground samples using a modified SS-ISO 13878 method and LECO CN-200 analyser. Treatment residue total N was analysed in the same way as larvae, but an additional NH₄-N measurement was performed on a wet sample using a FOSS TECATOR FIAstar 5000 Analyzer and following a modified method supplied in instrument application notes (AN 65/84, ASN 01/92 and ASN 02/92) and added to the total N value. The maturity of treatment residues was measured after adjusting its DM content to 28% (on average) by addition of dry sawdust (93% DM), using the Solvita® test following the procedure provided by the supplier.

2.5. Gas sampling

Emissions of GHG (CO₂, CH₄ and N₂O) and of NH₃ were sampled using a closed chamber technique. Each treatment box and a

battery-driven fan (Rubicson, Kjell & Co Elektronik AB, Sweden) were placed in a chamber closed with an airtight lid, equipped with a sampling port and pressure normalisation loop tube (1 m long, Ø 4 mm) for approximately 60 min. Frequency of gas sampling (Fig. 1) was selected to balance the ability to perform multiple treatments in triplicates and provide a sufficient resolution for gas emission dynamics. The gas samples were taken at intervals of 0, 15, 30 and 60 min using a 60 mL syringe and transferred to gas-tight N-evacuated 20 mL glass vials in pairs, which were later analysed by gas chromatography (GC) (Clarus 500, Perkin Elmer, USA) for CO₂, CH₄ and N₂O following the procedure described in Ermolaev et al. (2015). At each interval, after taking the two gas samples for GC analysis, NH₃ gas concentration was measured directly by connecting the opening of a NH₃ gas detection tube (105SD, 105SC, Kitagawa, Japan) to the sampling port and extracting the sample using a Kitagawa gas pump (AP-20, Kitagawa, Japan). The volume of extracted gas was recorded, as it resulted in chamber gas dilution through the pressure normalisation tube. This was compensated for by calculating the gas concentration using the known concentrations in the sample and room air and the known volumes of the chamber and extracted samples, described as:

$$PPM_{mix} = \frac{PPM_{orig} * V_{orig}}{V_{orig} + V_{added}} + \frac{PPM_{added} * V_{added}}{V_{orig} + V_{added}} \quad (1)$$

where PPM_{mix} is the actual measured concentration, PPM_{orig} is the original concentration before dilution and PPM_{added} is the concentration in the room air. V_{orig} and V_{added} represent volume of the sampling chamber and gas entering through the normalisation tube, respectively. The dilution at each sampling interval was also considered for the samples of the following intervals, by assuming a cumulative effect.

2.6. Data analysis

The concentrations measured at the intervals were used to calculate the emission rates in g min⁻¹ for all gases, using the slope estimation in the linear regression (LINEST function in Microsoft Excel 2016). This produced CO₂ emission rate values with a high level of accuracy (R² > 0.99, n = 18), confirming linearity of gas concentration increases during all sampling periods. Due to very low concentrations of CH₄, often close to background level, if a negative slope was obtained in linear regression of the four interval measurements, the emission rate was assumed to be zero. Due to even lower emissions of N₂O, and since N₂O concentration did not increase in the interval measurements in the majority of cases, it was assumed that N₂O emission rate was zero if R² was below

0.9. Based on the calculated gas flows, total emissions of all gases were estimated using trapezoidal rule integration of emission rates over time between sampling points (Holman, 2001), and assuming that the emission rates were constant from the start of treatment up to the first gas measurement and from the last gas measurement to the end of treatment. Such assumption, while imprecise, was necessary to make an estimation of total emissions and it was used to avoid underestimating the total emissions. Analysis of variance (ANOVA followed by the Tukey's HSD tests) with 95% confidence interval was used to verify statistically significant differences between the inflow material and treated material and between treatments. For comparisons of inflow material treated material and larvae produced, variables such as bioconversion ratio and material reduction were calculated on dry matter basis as previously described by Lalander et al. (2019). All statistical analyses were conducted in R (R Core Team, 2016).

The total GHG emissions were evaluated using GWP values of 34 and 298 for CH₄ and N₂O, respectively, over 100 years according to IPCC (2013).

3. Results

3.1. Material reduction and bioconversion ratios

The conversion efficiency was similar for all treatments, but with a slightly greater average mass reduction (52.3 ± 4.2) and larvae bioconversion ratio (27.9 ± 3.7) in the bacteria and larvae (BL) treatment (Table 1). However, there was high variation between replicates and no significant difference between treatments was found. Similarly, no significant difference between the treatments was found for any of the parameters analysed. The residues remaining after the treatments had a DM content of 22% on average, which was lower than that of the initial food waste (25%) (Table 1). Larval size differed slightly between the treatments, but the differences were not significant (Table 1). Larval survival estimation was based on initial number of larvae extrapolated from a smaller sample and the final number of larvae enumerated in their totality. The final number of larvae exceeded the initial estimation by 22% in one of the replicates and by 1% in another in treatment BL. In those cases, the survival was assumed to be 100% and the initial larvae numbers were adjusted to 856 and 710, respectively, bringing the average survival in that treatment to 98.3%. With such an adjustment, the larval survival in treatment BL was significantly higher than in treatment PBL, although with a relatively high P value of 0.05 (Table 1).

Residue maturity was low across the treatments, with a compost maturity index (CMI) of 2.3 on average ($n = 9$), indicating that

the residues could be classified as very active compost (Table 1). While CMI is calculated based on Solvita[®] measurements of CO₂ and NH₃ emissions, in our treatments these measurements generally showed no detectable NH₃ ($n = 7$) and high CO₂ concentrations (data not shown).

Overall, the majority of available DM and VS remained in the products after the treatment, and 67–78% of initial dry solids and VS available in the substrate was recovered in the larvae and residues after treatment (Table 2). Most of the water initially available in the material was recovered in the residues (Table 2), resulting in relatively low residue DM (Table 1). Larvae contained 10–17% of the initial water, while 18–31% of water was lost during the treatment (cf. Table 2). As water is also generated in the process of respiration, it was possible to estimate the amount of water produced based on the VS loss. On average, 64.3 g VS was lost during the larvae composting in treatments L, BL and PBL, while an additional 30.3 g VS was lost, on average, in the pre-treatment of PBL (cf. Table 2). Based on the oxidation reaction of the average food waste formula (Haug, 1993) it was estimated that, on average, 34.2 g of water were generated during the larvae composting step in all treatments and an additional 16.1 g during pre-treatment of PBL due to VS loss in respiration.

Larvae contained a higher amount of N ($60.7 \text{ mg N g}^{-1} \text{ DM}$, on average) than the food waste substrate (Table 2), while the residues contained a lower amount. However, from a mass balance perspective, over a half the initial N ended up in larvae and the rest was recovered in the residues (Table 2). Total N in all treatment products was close to 100% of the initial N, suggesting high N recovery, however most likely also caused by variation between the replicates and possible underestimation of initial N amount in the food waste. Variation associated with analytical methods and process development in pre-treatments and treatments in some cases resulted in overestimation in N balance due to counter-active combinations of such variations.

3.2. Gas emissions

The CO₂ emission rates and patterns were similar during the larvae composting step in the different treatments. The CO₂ emission rate peaked during the second measurement on day 7 of fly larvae composting, reaching 7.4 mg min^{-1} on average across the treatments, and then declined slightly by day 11 to 5.6 mg min^{-1} (Fig. 2A). Total emissions of CO₂ were similar across all treatments, excluding any emissions from pre-treatment (Fig. 2B). Total emissions of CO₂ in the larvae treatment step were on average $96.5 \text{ g CO}_2 \text{ kg}^{-1}$ initial food waste on a wet weight (WW) basis (Fig. 2B, Table 3).

Table 1
Properties of larvae and residues after 14 days of treatment of food waste in three fly larvae composting treatments (arithmetic mean $\pm$ standard deviation, $n = 3$). DM = dry matter, WW = wet weight, VS = volatile solids, init. = initial.

Property	Treatment		
	Larvae (L)	Bacteria and larvae (BL)	Bacteria pre-treatment, larvae (PBL)
Larvae survival ¹ , %	87.0 ± 12.2	98.3 ± 2.4	67.4 ± 12.6
Final larvae weight, mg larvae ⁻¹	297 ± 4	307 ± 3	309 ± 9
Larvae DM, % of WW	34.3 ± 1.6	35.7 ± 0.4	36.4 ± 0.2
Larvae VS, % of DM	90.2 ± 0.6	90.9 ± 0.4	91.4 ± 0.6
Residue DM, % of WW	21.9 ± 1.9	19.3 ± 1.7	22.2 ± 1.5
Residue VS, % of DM	90.3 ± 1.2	89.1 ± 1.2	88.9 ± 1.1
Mass reduction, % of init. DM	43.0 ± 7.3	52.3 ± 4.2	48.9 ± 6.7 (43.2 ± 6.1) ²
Bioconversion ratio, % of init. DM	21.3 ± 3.4	27.9 ± 3.7	17.9 ± 3.8 (20.1 ± 4.6) ²
Residue pH	4.7 ± 0.2	5.1 ± 0.2	5.4 ± 0.3
Residue Solvita [®] index	2.7 ± 0.5	2.3 ± 0.5	2.0 ± 0.8

¹ Estimated based on weighing a sample of larvae, so a value $>100\%$ is possible.

² Calculated from start of larvae treatment, i.e. excluding reduction in pre-treatment.

Table 2

Mass balances based on water, dry matter (DM) and volatile solids (VS) content for the treatments with fly larvae (L), bacteria and larvae treatment (BL) and bacteria pre-treatment and larvae treatment (PBL), and total nitrogen (N) balance expressed as amount of N measured per g DM and as percentage of initial nitrogen (arithmetic mean $\pm$ standard deviation in all cases).

	Food waste ¹	Larvae		Residue		Total in products
DM	g	g	% of initial DM	g	% of initial DM	% of initial DM
L	291.7 $\pm$ 1.7	62.2 $\pm$ 10.2	21 $\pm$ 3	166.3 $\pm$ 20.4	57 $\pm$ 7	78
BL	292.8 $\pm$ 0.0	81.6 $\pm$ 10.8	28 $\pm$ 4	139.6 $\pm$ 12.4	48 $\pm$ 4	76
PBL	297.8 $\pm$ 1.2 (267.0 $\pm$ 9.8) ²	53.2 $\pm$ 11.1 (53.2 $\pm$ 11.1) ²	18 $\pm$ 4 (20 $\pm$ 5) ²	152.2 $\pm$ 20.5 (152.2 $\pm$ 20.5) ²	51 $\pm$ 7 (57 $\pm$ 6) ²	69 (77) ²
Water	g	g	% of initial water	g	% of initial water	% of initial water
L	880.5 $\pm$ 5.1	118.4 $\pm$ 13.5	13 $\pm$ 1	589.5 $\pm$ 11.5	67 $\pm$ 2	80
BL	883.9 $\pm$ 0.1	146.9 $\pm$ 16.6	17 $\pm$ 2	581.8 $\pm$ 12.1	66 $\pm$ 1	82
PBL	899.1 $\pm$ 3.5 (786.8 $\pm$ 28.9) ²	93.1 $\pm$ 19.3 (93.1 $\pm$ 19.3) ²	10 $\pm$ 2 (12 $\pm$ 3) ²	531.8 $\pm$ 31.0 (531.8 $\pm$ 31.0) ²	59 $\pm$ 3 (68 $\pm$ 4) ²	69 (80) ²
VS	g	g	% of initial VS	g	% of initial VS	% of initial VS
L	269.2 $\pm$ 1.5	56.2 $\pm$ 9.5	21 $\pm$ 3	150.3 $\pm$ 20.4	56 $\pm$ 8	77
BL	270.2 $\pm$ 0.0	74.2 $\pm$ 9.9	27 $\pm$ 4	126.2 $\pm$ 12.8	47 $\pm$ 5	74
PBL	274.9 $\pm$ 1.1 (244.6 $\pm$ 9.0) ²	48.6 $\pm$ 10.1 (48.6 $\pm$ 10.1) ²	18 $\pm$ 4 (20 $\pm$ 5) ²	135.5 $\pm$ 19.9 (135.5 $\pm$ 19.9) ²	49 $\pm$ 7 (55 $\pm$ 7)	67 (75)
Total N	mg N g ⁻¹ DM	mg N g ⁻¹ DM	% of initial N	mg N g ⁻¹ DM	% of initial N	% of initial N
L	63.2 $\pm$ 3.7	58 $\pm$ 5	58 $\pm$ 5	24.0 $\pm$ 1.3	38 $\pm$ 3	97
BL	35.4 $\pm$ 2.9	55.7 $\pm$ 2.5	57 $\pm$ 5	23.1 $\pm$ 1.1	43 $\pm$ 4	100
PBL		63.4 $\pm$ 2.8	61 $\pm$ 8	29.6 $\pm$ 1.7	32 $\pm$ 5	93

¹ Total N value for food waste taken from a previous study (Lalander et al., 2019) performed on the same batch of food waste mixture.

² Calculated from start of larvae treatment, i.e. excluding changes occurring in pre-treatment.

The CO₂ emissions during the pre-treatment step in the PBL treatment were estimated from VS loss as suggested by Haug (1993), assuming a carbon fraction of 55% of VS. This resulted in an additional 51.5 $\pm$ 13.7 g CO₂ kg⁻¹ initial food waste (WW) being produced on average in the pre-treatment step, assuming that all the carbon in the lost VS was converted to CO₂.

In all three treatments, the CH₄ emission rates were relatively low and they generally declined from start to end of larvae composting (Fig. 2C). Initial CH₄ emissions at the beginning of larvae composting in treatment PBL were lower on average than in the experiments without pre-treatment and were closer to CH₄ emission rates measured on day 11 of those experiments (Fig. 2C). This difference was not significant for any of the sampling days, but when comparing the emission rates grouped by day and treatment, the CH₄ emission rates in treatment PBL were significantly lower than those in treatment BL ($P = 0.005$). The lower CH₄ emission rates resulted in lower overall total CH₄ emissions (Fig. 2D, Table 3). Total emissions differed between treatments and were significantly lower in the larvae composting step of the PBL treatment with pre-treated substrate ($P = 0.02$) than in treatment BL, where bacteria and larvae treated the substrate (Fig. 2D). The emissions did not differ significantly between treatments L and BL or between treatments L and PBL (Fig. 2D). The CH₄ emissions were not measured during the pre-treatment step and were not included in the comparisons, but they could potentially increase the total CH₄ emissions from the whole PBL treatment process.

The N₂O emissions did not show any specific pattern and the majority of the values measured were very close to background levels (Fig. 2E). The flux values were often assumed to be zero, as the interval measurements in the chamber produced low R² values ($R^2 < 0.9$, $n = 9$ of 27) due to both initial N₂O concentration close to background levels and no concentration increase in the chamber over the measuring period. The N₂O results presented here are thus an estimate of the actual emissions.

The estimated N₂O emission rates varied between replicates and thus the mean values had relatively high standard deviation (Table 3). The emission rates during larvae composting in treatment PBL were so low that they were not significantly different from zero (Fig. 2E). This resulted in cumulative N₂O emissions in

this treatment being very small, and the standard deviation being comparable to the mean of the replicates (Fig. 2F, Table 3). Overall, cumulative N₂O emissions from the larvae composting step were not significantly different between the three treatments and showed large variation. The N₂O emissions from the pre-treatment step in treatment PBL were not measured.

Ammonia emissions were below the detection limit of 0.2 ppm in all cases ($n = 27$).

4. Discussion

4.1. Process efficiency

Bioconversion ratio and substrate reduction were not significantly different between the treatments, as evidenced by the similarity in total CO₂ emissions between the treatments, indicating that a similar amount of the VS content in the substrate was converted to larval biomass (Table 1, Fig. 2A–B). Mean estimated total CO₂ emissions in all treatments were 96.5 g CO₂ kg⁻¹ initial food waste (WW basis), corresponding to a loss of 53.8 g VS (based on carbon conversion), which was less than the overall average of 64.3 g VS estimated to be lost from the mass balance (Table 2). This difference could be explained by either underestimation of the total CO₂ emissions, emissions of other volatile substances not measured in this study or inaccuracy in VS measurements.

Overall, after larvae composting in all treatments, approximately half of the initial VS was recovered in the residues (Table 2). The relationship between ash, VS and DM only changed by 2–5 percentage units, from around 92 to 98%, as some minerals were taken up by the larvae together with VS. The high proportion of initial VS remaining in the residues suggests that, even after larval composting in all treatments, the residues could still be considered only mildly degraded (Table 2). This was confirmed by the low Solvita[®] CMI value obtained for the residues (Table 1). This is typical for fly larvae composting of food waste, as BSF larvae grow quickly and thus consume the most easily digestible parts of the substrate (Li et al., 2011; Li et al., 2015). The part of VS converted by the larvae (representing bioconversion ratio on a VS basis) was 21 and 28% of initial VS in treatments L and BL, respectively (Table 1),

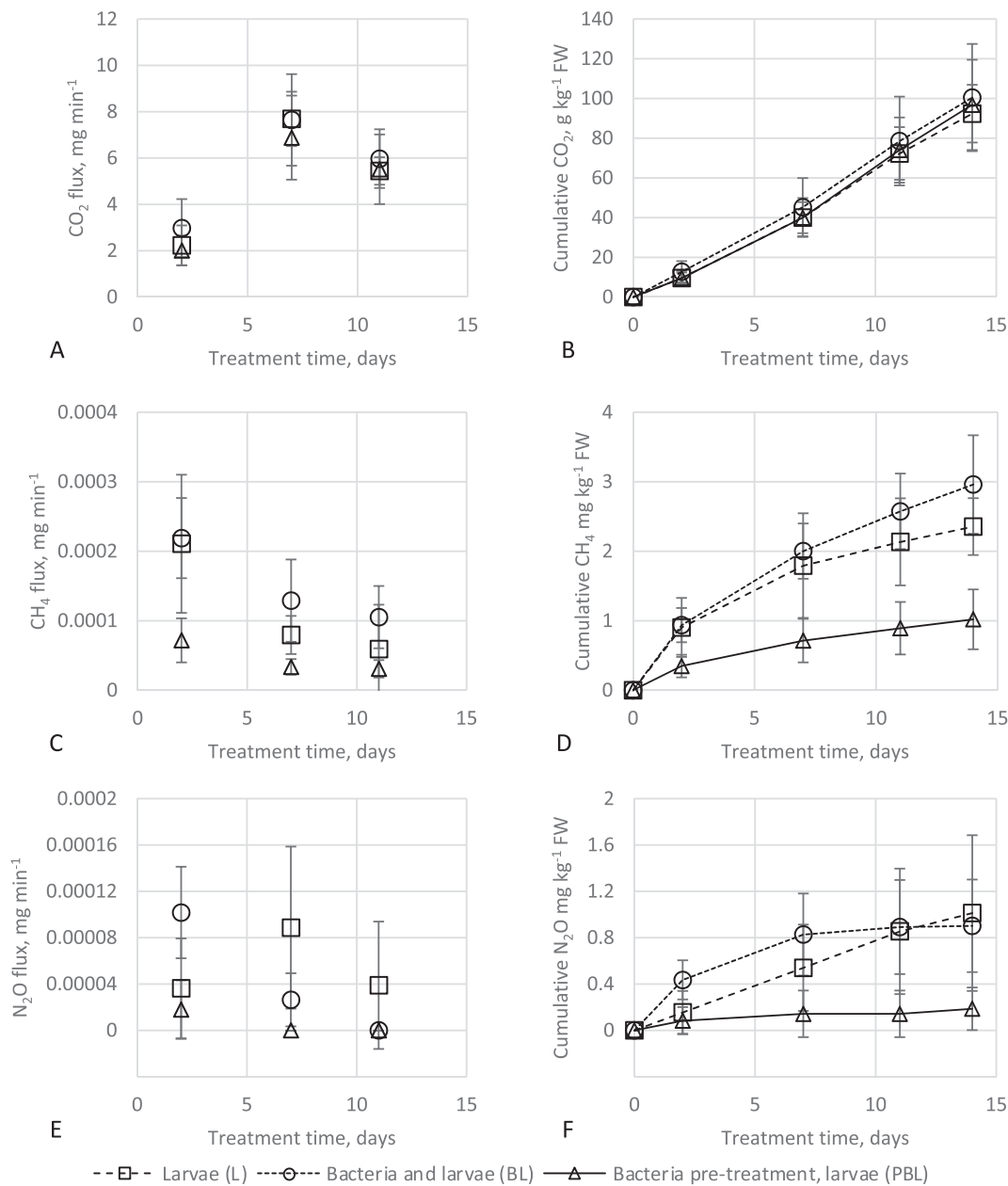


Fig. 2. (A) Carbon dioxide (CO₂) emission rates, (B) estimated cumulative CO₂ production, (C) methane (CH₄) emission rates, (D) estimated cumulative CH₄ production, (E) nitrous oxide (N₂O) emission rates and (F) estimated cumulative N₂O production expressed per kg wet food waste (FW) during the larvae composting step in all treatments. Whiskers show standard deviation for replicates (n = 3).

Table 3
Cumulative emissions of CO₂, CH₄ and N₂O during the larvae composting step in the treatments with fly larvae (L), bacteria and larvae treatment (BL) and pre-treatment with bacteria and larvae treatment (PBL), expressed per kg of wet food waste (FW), and emissions of CH₄ and N₂O expressed as g CO₂-equivalents per kg FW, calculated based on their global warming potential (GWP) over 100 years. Mean ± standard deviation (n = 3).

	Total emissions, kg ⁻¹ FW			Emissions of CH ₄ and N ₂ O, g CO ₂ -eq. kg ⁻¹ FW
	CO ₂ (g)	CH ₄ (mg)	N ₂ O (mg)	g CO ₂ -eq.
L	92.4 ± 14.5	2.35 ± 0.41	1.013 ± 0.672	0.382 ± 0.187
BL	100.5 ± 27.0	2.96 ± 0.71	0.902 ± 0.400	0.376 ± 0.095
PBL	96.8 ± 22.7 ¹	1.02 ± 0.43 ¹	0.186 ± 0.184 ¹	0.090 ± 0.045 ¹

¹ Calculated based only on emissions from larvae treatment, i.e. excluding pre-treatment.

which is similar to the conversion ratio of 35% of initial DM reported previously for food waste (Lalander et al., 2018). Bioconversion ratios reported for other substrates indicate that food

waste represents a good substrate for rearing black soldier fly larvae. Bioconversion ratios can depend on factors such as feed composition and feeding rate when other process parameters are

optimal. The effect of the feeding rate is e.g. seen in the study by Lalander et al. (2019) where a lower bioconversion ratio of 13.9% of DM was achieved for food waste with higher feeding rate of 40 mg DM larvae⁻¹ d⁻¹, suggesting that a feeding rate of 30.0 mg DM larva⁻¹ d⁻¹ selected in current study resulted in a better conversion efficiency.

Overall, the nitrogen balance indicated that, of the initial N in the food waste, over a half was recovered in the larvae and the rest in the treatment residues (Table 2). In treatment L, 97% of the initial N in the substrate was recovered in these two fractions, while in treatment BL 100% was recovered. This indicates small N losses from the system. In treatment PBL, the high N recovery in the two fractions produced suggests that there was no major N loss either during the pre-treatment step or during larvae composting. Two major factors were identified as reasons for the very high N content in the fractions produced in relation to the initial N in all treatments (Table 2). The primary factor was probably underestimation of total N value in the incoming food waste, as the initial sampling may not have reflected the total composition of the material due to unexpected inhomogeneity of incoming material. In fact, another study performed on food waste from the same source reported a higher initial N value (49.5 mg N g⁻¹ DM; unpublished data) than found here (35.4 mg N g⁻¹ DM; Table 2). The second factor was deviation in substrate analysis (the method has a 10% error). The N content of the larvae has been shown to be consistent within a certain range for BSF larvae and does not depend strongly on initial N, as long as it is within an acceptable range for their growth (Lalander et al., 2019).

4.2. Gas emissions

The CO₂ emissions were similar between the treatments throughout the study and showed a similar pattern, with higher CO₂ emission rate measured on day 7 of the experiments and a slight decrease on day 11 (Fig. 2A). This emissions pattern differs from that in the thermophilic composting process, where an initial peak is typically observed early in the microbial degradation process, when easily available substances are consumed by the proliferating microorganisms (Ermolaev et al., 2015). A second peak associated with process temperature increase if the compost becomes thermophilic, and with more substances becoming easily degradable at such temperatures, can be expected (Ermolaev et al., 2015).

In fly larvae composting, the substrate development had a lower effect on the CO₂ emissions (Fig. 2A–B). Furthermore, in treatment PBL with pre-treated substrate, there was no significant difference in CO₂ produced during larvae treatment compared with the other two treatments, suggesting the larvae themselves are the major source of emissions, since the bacteria introduced in pre-treatment (PBL) and seeded at start of the treatment (BL) would contribute to CO₂ emissions due to their respiration in addition to that of larvae themselves (Table 3). The near linearity in cumulative CO₂ emissions could also be expected, as the larvae grow in a nearly linear way (Lalander et al., 2019) and thus their respiration could be expected to increase accordingly (Banks, 2014). No strong correlation ($R^2 = 0.53$) between CO₂ emission rates and larvae size was observed in our experiments, but it was significant ($P < 0.0001$) and in the cases where larvae were small, the emission rates were also smaller, while the CO₂ flux was never below 0.003 g min⁻¹ after the larvae reached a size of 0.2 g (Fig. 3). This means that while there was a significant correlation with larvae size, it could not explain all the variance in CO₂ production rate. Larger variation in CO₂ emission rates later in the process can be related to multiple factors, including the establishment of a microbial community, resulting in a larger contribution of microbial res-

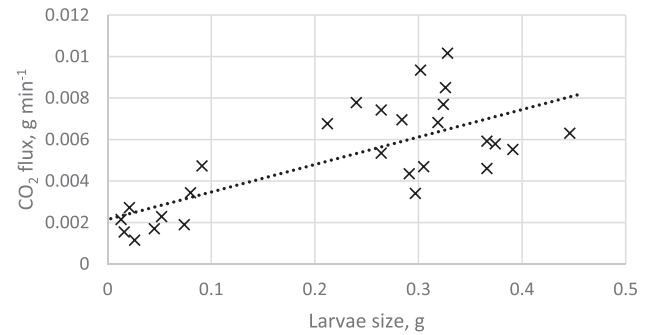


Fig. 3. Change in CO₂ emission rates with increasing larvae size in all treatments. All data points (x) for treatments L, BL and PBL and their replicates are shown ($n = 27$). The dotted line shows the regression function: CO₂ flux, g min⁻¹ = 0.002 + 0.013 × Larvae size, g ($R^2 = 0.53$, $P < 0.0001$).

piration to total CO₂ emissions and differences in larval survival between treatments and replicates (Table 1).

The CH₄ emissions dynamics suggested that during the process of larvae treatment, the material becomes less anaerobic, thus reducing CH₄ production rates during larvae composting (Fig. 2C). There was only a weak correlation ($R^2 = 0.43$, 0.42 and 0.38 for treatments L, BL and PBL, respectively) between CH₄ emission rates and treatment time, mostly due to variation in CH₄ flux measured in different replicates (Fig. 2C–D). However, parts of the substrate remained anaerobic as CH₄ production continued, although at much slower rate on days 7 and 11 (Fig. 2C–D). In aerobic composting, the CH₄ emissions typically increase 4–10 days later in the process, due to a combination of factors, including a longer time required for methanogen population growth and thermophilic temperatures (Ermolaev et al., 2015; Ermolaev et al., 2012; Jäckel et al., 2005). In our fly larvae composting treatments the opposite was observed. The reduction in CH₄ emissions with treatment time could be associated with larvae growth, as they turn material more efficiently when they are larger and thus aerate it more (Cickova et al., 2015).

Overall, the total CH₄ emissions from the treatment period were similar between the two treatments without pre-treatment, with no significant difference, while these emissions were significantly smaller in treatment PBL compared with BL (Table 3). The total CH₄ emissions were very small, one-thousandth of the level observed during conventional composting (Ermolaev et al., 2012). This likely relates to the majority of the process being aerobic, performed in thin layers and constantly turned. When compared on a carbon basis, the total CH₄ emissions represented 3.3 g of VS loss (5.3% of total VS), only a fraction of the carbon emissions compared with CO₂, which was calculated to be responsible for losses of 53.8 g VS (87%) on average during all treatments.

The CH₄ emissions from pre-treatment in PBL were not measured. While it was possible to estimate the emissions of CO₂ from VS loss due to degradation during the pre-treatment period (Table 3) with relatively high certainty, the CH₄ emissions were more difficult to predict. Two possibilities were considered: (1) No CH₄ was emitted during pre-treatment, which was supported by the results obtained for conventional composting, when there is typically a delay in peak CH₄ production of 4–10 days (Ermolaev et al., 2015; Ermolaev et al., 2012) and production is associated with thermophilic temperatures (Jäckel et al., 2005), which were not reached during pre-treatment in our experiments. (2) The emission rate during pre-treatment was similar to that measured during the first sampling in the PBL treatment, in which case the total emissions during the 7 days of pre-treatment were an estimated 0.61 ± 0.27 mg CH₄ kg⁻¹ FW, bringing the total CH₄

emissions in treatment PBL to 1.63 ± 0.70 mg CH₄ kg⁻¹ FW, a value still lower than in the other treatments (Table 3).

The results obtained indicate very low (or no) N₂O and NH₃ emissions, as also indicated by the total N recovery in the N balance (Table 2, Fig. 2E–F). A lack of detectable NH₃ emissions was confirmed by the Solvita® measurements (Table 1). The lowest reading, 4.5 out of 5 units, was detected only from residues in one out of nine replicates in all treatments. In a study by Lalander et al. (2015), NH₃ emissions from fly larvae composting were detected (olfactory identification) and confirmed through N balance, indicating a loss of up to 50% of initial N through emissions to air. Moreover, that study reported higher pH in the residues, which would enhance the NH₃ emissions. The low NH₃ emissions in the present study could be dependent on several factors. First, the final residue pH was 5.1 on average (Table 1). Second, there may be a lack of NH₄⁺ in the residues, as also observed by Green and Popa (2012), which is likely due to larval assimilation of N-containing molecules such as proteins. On the other hand, Cickova et al. (2015) suggest that in fly larvae composting, substantial NH₃ emissions can be expected from the increasingly aerobic process due to larval movement in the substrate. Those authors also suggest that the material can dry out or warm up and the pH of the residues can become alkaline, which was not the case in the present study (see Table 1). Third, other studies reporting high ammonia emissions from fly larvae treatment have used manure as substrate (Cickova et al., 2012; Iñiguez-Covarrubias et al., 1994), and manure can be expected to contain higher concentrations of degraded N forms and generally have a higher pH. These may be associated with microbial degradation of remaining organic N and other forms of N excreted by larvae due to their metabolism, such as uric acid (Bursell, 1967). The importance of moisture and pH changes should be investigated in future experiments, to extend knowledge on NH₃ emission dynamics during fly larvae composting. In conventional composting, it has been reported that factors such as process temperature becoming thermophilic and pH increasing above 6 are responsible for the NH₃ emissions (Ermolaev et al., 2015).

The residues in all treatments were not stabilised during larvae composting (Table 1) and the Solvita® test indicated stability similar to that of fresh compost. Thus the treatment residues require an additional treatment step for stabilisation. Previous studies have shown that *Salmonella* spp. and viruses (reoviruses and adenoviruses) are inactivated in the larvae composting process, but not parasites (*Ascaris suum*) (Lalander et al., 2013; Lalander et al., 2015). During additional treatment steps to stabilise and sanitise the residue, there is a risk of additional emissions, and thus the optimal conditions for emissions reduction have to be maintained and emissions monitored. In a full-scale environmental assessment, management of the residues needs to be included.

4.3. Perspectives

Fly larvae composting not only performs a valuable function in the form of organic waste management, but also generates a valuable product in the form of larval biomass (Lalander et al., 2018). When the emissions from a waste treatment alternative are being compared, a comprehensive systems analysis should be performed, where results from multiple studies are incorporated. Such a study should consider the value of larvae biomass as the product. In such cases, emissions per unit mass of larvae and per ton of dry feed replacement produced can be estimated (Table 4). This study focused on the direct emissions from the treatment step. The CO₂ equivalents of emissions of CH₄ and N₂O were 1/1000 of the actual CO₂ emissions when expressed per ton of dry larval biomass produced (Table 4). Treatment PBL produced less CH₄ and N₂O than the other two treatments, but total emissions of CO₂ nearly doubled, which resulted in significantly higher total emissions per ton DM of larvae produced including CO₂, CH₄ and N₂O in treatment PBL comparing to L and BL ($P < 0.00005$). However, this did not result in a significant increase in bioconversion ratio, indicating that such pre-treatment is unnecessary (Tables 2 and 5). A similar conclusion can be drawn when looking at emissions per 1000 larvae also resulting in significantly higher ($P < 0.0005$) total emissions in treatment PBL comparing to L and BL (Table 4). A complete system evaluation should thus complement the results obtained in this study with other relevant impacts and when other feeds are replaced with larvae produced.

Evaluation of direct emissions from fly larvae composting enables comparison of this treatment with treatment steps in other common organic waste treatment alternatives, such as composting and anaerobic digestion (Table 5). Emissions of CO₂ during fly larvae composting are of biogenic origin and may be considered non-

Table 5

Comparison of direct emissions from treatment of organic food wastes (per ton of wet food waste (FW) treated) using common methods. CO₂-equivalents are calculated based on global warming potential of the respective gases over 100 years (IPCC, 2013).

Treatment method	Total emissions ton ⁻¹ FW		
	Biogenic CO ₂ , kg	CH ₄ , kg CO ₂ -eq.	N ₂ O, kg CO ₂ -eq.
Fly larvae composting (current study)	96	0.09	0.29
Fly larvae composting (Mertenat et al., 2019)	–	0.01	2.5
Windrow composting ¹	144	18	1.2
Anaerobic digestion	50–84 ²	0–48 ²	0–23 ³

¹ Average value from Ermolaev et al. (2012) excluding emissions from compost post-treatment.

² Averages estimated based on the ranges given by Möller et al. (2009) excluding emissions from digestate handling and biogas upgrading and use.

³ According to Thanh Phong et al. (2016) due to emissions from biofilter.

Table 4

Estimated direct emissions of CH₄ and N₂O in g CO₂-equivalents, assuming global warming potential (GWP) values over 100 years, and of CO₂ in tons per ton of dry matter (DM) larvae biomass produced and per 1000 larvae produced in 14 days in the treatments with fly larvae (L), bacteria and larvae treatment (BL) and pre-treatment with bacteria and larvae treatment (PBL). Mean $\pm$ standard deviation ($n = 3$).

	Per ton DM of larvae produced			Per 1000 larvae produced		
	CO ₂	CH ₄ and N ₂ O	Total incl. CO ₂ , CH ₄ and N ₂ O	CO ₂	CH ₄ and N ₂ O	Total incl. CO ₂ , CH ₄ and N ₂ O
	ton	kg CO ₂ -eq.	ton CO ₂ -eq.	g	mg CO ₂ -eq.	g CO ₂ -eq.
L	1.75 $\pm$ 0.17	7.90 $\pm$ 4.73	1.76 $\pm$ 0.17	178.7 $\pm$ 19.0	788 $\pm$ 458	179.5 $\pm$ 18.9
BL	1.42 $\pm$ 0.19	5.53 $\pm$ 1.98	1.43 $\pm$ 0.19	156.4 $\pm$ 24.1	602 $\pm$ 206	157.0 $\pm$ 24.0
PBL	3.08 $\pm$ 0.07 ¹	3.97 $\pm$ 3.43 ¹	3.08 $\pm$ 0.07 ¹	345.2 $\pm$ 3.2 ¹	280 $\pm$ 150 ¹	345.5 $\pm$ 3.1 ¹
	(1.91 $\pm$ 0.16) ²	(2.10 $\pm$ 1.54) ²	(1.91 $\pm$ 0.15) ²	(214.5 $\pm$ 22.2) ²	(231 $\pm$ 163) ²	(214.8 $\pm$ 22.1) ²

¹ Including estimated emissions in the pre-treatment step based on VS loss for CO₂ and based on first gas measurement for CH₄ and N₂O.

² Calculated based only on emissions from the larvae treatment step, i.e. excluding pre-treatment.

contributing to global climate change (Bernstad and Jansen, 2012). In such an interpretation of total GHG emissions from fly larvae composting, the total emissions are very small, only around 0.38 kg CO₂-eq. ton⁻¹ food waste treated (Table 3). A recent study by Mertenat et al. (2019) on emissions of CH₄ and N₂O from larger crate fly larvae composting reported lower total CH₄ and higher N₂O emissions (Table 5). However, the N₂O emissions pattern was different from that in the present study, with emission peaks appearing after feeding events (Mertenat et al., 2019). In the present study, the gas measurements were performed less frequently and before the feeding events, to avoid overestimation of the emissions associated with system disturbance due to feed addition. Similar N₂O emission peaks to those appearing directly after feed addition have been observed in composting, at the start of the process, if the food waste contains nitrate (Ermolaev et al., 2015). This suggests that further investigation may be necessary, with more frequent gas measurements and nitrate content evaluation of incoming food waste.

Organic waste treatment alternatives often include several process steps that may contribute additional direct GHG emissions to the overall system total. In anaerobic digestion, additional emissions can be expected due to biofilter emissions and digestate storage and during upgrading and combustion of biogas (Møller et al., 2009). Since an additional treatment step is necessary for fly larvae composting residues, additional NH₃ or N₂O emissions can be expected if the residues are still relatively moist, similarly to possible emissions during the curing stage of the composting process (Ermolaev et al., 2012).

The emissions from fly larvae composting residue post-treatment have not yet been evaluated. Direct and indirect emissions during all the process stages for fly larvae composting have to be performed to allow fair comparisons.

5. Conclusions

Overall GHG emissions from fly larvae composting of food waste using BSF larvae were comparatively low. CO₂ from food waste degradation was the major GHG emitted, with total emissions of 96 g CO₂ per kg food waste treated. Emissions of CH₄ and N₂O were equivalent to 0.38 kg CO₂-equivalents per ton food waste treated (assuming GWP over 100 years). No NH₃ emissions were detected, which was reflected in total N mass balance over the treatment cycle.

There was no significant improvement in material reduction or bioconversion ratio when food waste was seeded with larvae-associated bacteria, compared with food waste treatment with larvae alone. Moreover, there was no significant effect of different treatments on total GHG emissions.

Acknowledgements

We would like to thank Frida Erlöv for her contribution to this project. The study was part of the Vinnova-funded project Upscaling fly larvae composting as a treatment method in Sweden (Registration number 2017-03210).

References

Allegretti, G., Talamini, E., Schmidt, V., Bogorni, P.C., Ortega, E., 2018. Insect as feed: An energy assessment of insect meal as a sustainable protein source for the Brazilian poultry industry. *J. Clean. Prod.* 171, 403–412. <https://doi.org/10.1016/j.jclepro.2017.09.244>.

Banks, I.J., 2014. To assess the impact of black soldier fly (*Hermetia illucens*) larvae on faecal reduction in pit latrines. *London School Hygiene Tropical Med.* <https://doi.org/10.17037/PUBS.01917781>.

Bernstad, A., Jansen, J.L., 2012. Review of comparative LCAs of food waste management systems - current status and potential improvements. *Waste Manage.* 32, 2439–2455. <https://doi.org/10.1016/j.wasman.2012.07.023>.

Bursell, E., 1967. The excretion of nitrogen in insects. In: Beament, J.W.L., Treherne, J.E., Wigglesworth, V.B. (Eds.), *Adv. Insect Physiol.*. Academic Press, pp. 33–67. [https://doi.org/10.1016/S0065-2806\(08\)60207-6](https://doi.org/10.1016/S0065-2806(08)60207-6).

Cickova, H., Newton, G.L., Lacy, R.C., Kozanek, M., 2015. The use of fly larvae for organic waste treatment. *Waste Manage.* 35, 68–80. <https://doi.org/10.1016/j.wasman.2014.09.026>.

Cickova, H., Pastor, B., Kozanek, M., Martinez-Sanchez, A., Rojo, S., Takac, P., 2012. Biodegradation of pig manure by the housefly, *Musca domestica*: a viable ecological strategy for pig manure management. *PLoS One* 7. <https://doi.org/10.1371/journal.pone.0032798>.

Diener, S., Solano, N.M.S., Gutierrez, F.R., Zurbrugg, C., Tockner, K., 2011. Biological treatment of municipal organic waste using black soldier fly larvae. *Waste Biomass Valor.* 2, 357–363. <https://doi.org/10.1007/s12649-011-9079-1>.

Ermolaev, E., Jarvis, A., Sundberg, C., Smårs, S., Pell, M., Jönsson, H., 2015. Nitrous oxide and methane emissions from food waste composting at different temperatures. *Waste Manage.* 46, 113–119. <https://doi.org/10.1016/j.wasman.2015.08.021>.

Ermolaev, E., Pell, M., Smårs, S., Sundberg, C., Jönsson, H., 2012. Greenhouse gas emission from covered windrow composting with controlled ventilation. *Waste Manage. Res.* 30, 155–160. <https://doi.org/10.1177/0734242X11426171>.

Green, T.R., Popa, R., 2012. Enhanced ammonia content in compost leachate processed by black soldier fly larvae. *Appl. Biochem. Biotechnol.* 166, 1381–1387. <https://doi.org/10.1007/s12010-011-9530-6>.

Haug, R.T., 1993. *The Practical handbook of compost engineering*. Lewis Publishers, Boca Raton, Florida, FL, USA. <https://doi.org/10.1201/9780203736234>.

Holman, J.P., 2001. *Experimental methods for engineers*. McGraw-Hill.

Iñiguez-Covarrubias, G., de Franco-Gómez, M.J., Andrade-Maldonado, G.d.R., 1994. Biodegradation of swine waste by house-fly larvae and evaluation of their protein quality in rats. *J. Appl. Anim. Res.* 6, 65–74. <https://doi.org/10.1080/09712119.1994.9706025>.

IPCC, 2013. *Climate Change 2013: The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA. <https://doi.org/10.1017/CBO9781107415324>.

IPCC, 2014. *Mitigation of Climate Change. Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA. <https://doi.org/10.1017/CBO9781107415416>.

Iwasa, M., Moki, Y., Takahashi, J., 2015. Effects of the activity of coprophagous insects on greenhouse gas emissions from cattle dung pats and changes in amounts of nitrogen, carbon, and energy. *Environ. Entomol.* 44, 106–113. <https://doi.org/10.1093/ee/nvu023>.

Jäckel, U., Thummes, K., Kämpfer, P., 2005. Thermophilic methane production and oxidation in compost. *FEMS Microbiol. Ecol.* 52, 175–184. <https://doi.org/10.1016/j.femsec.2004.11.003>.

Jarvis, A., Sundberg, C., Milenkovski, S., Pell, M., Smårs, S., Lindgren, P.E., Hallin, S., 2009. Activity and composition of ammonia oxidizing bacterial communities and emission dynamics of NH₃ and N₂O in a compost reactor treating organic household waste. *J. Appl. Microbiol.* 106, 1502–1511. <https://doi.org/10.1111/j.1365-2672.2008.04111.x>.

Jeon, H., Park, S., Choi, J., Jeong, G., Lee, S.-B., Choi, Y., Lee, S.-J., 2011. The intestinal bacterial community in the food waste-reducing larvae of *hermetia illucens*. *Curr. Microbiol.* 62, 1390–1399. <https://doi.org/10.1007/s00284-011-9874-8>.

Lalander, C., Diener, S., Magri, M.E., Zurbrugg, C., Lindstrom, A., Vinneras, B., 2013. Faecal sludge management with the larvae of the black soldier fly (*Hermetia illucens*) - from a hygiene aspect. *Sci. Total Environ.* 458, 312–318. <https://doi.org/10.1016/j.scitotenv.2013.04.033>.

Lalander, C., Diener, S., Zurbrugg, C., Vinneras, B., 2019. Effects of feedstock on larval development and process efficiency in waste treatment with black soldier fly (*Hermetia illucens*). *J. Clean. Prod.* 208, 211–219. <https://doi.org/10.1016/j.jclepro.2018.10.017>.

Lalander, C., Nordberg, A., Vinneras, B., 2018. A comparison in product-value potential in four treatment strategies for food waste and faeces - assessing composting, fly larvae composting and anaerobic digestion. *Glob. Change Biol. Bioenergy* 10, 84–91. <https://doi.org/10.1111/gcbb.12470>.

Lalander, C.H., Fidjeland, J., Diener, S., Eriksson, S., Vinneras, B., 2015. High waste-to-biomass conversion and efficient *Salmonella* spp. reduction using black soldier fly for waste recycling. *Agron. Sustain. Dev.* 35, 261–271. <https://doi.org/10.1007/s13593-014-0235-4>.

Li, Q., Zheng, L.Y., Qiu, N., Cai, H., Tomberlin, J.K., Yu, Z.N., 2011. Bioconversion of dairy manure by black soldier fly (Diptera: Stratiomyidae) for biodiesel and sugar production. *Waste Manage.* 31, 1316–1320. <https://doi.org/10.1016/j.wasman.2011.01.005>.

Li, W., Li, Q., Zheng, L., Wang, Y., Zhang, J., Yu, Z., Zhang, Y., 2015. Potential biodiesel and biogas production from corn cob by anaerobic fermentation and black soldier fly. *Bioresour. Technol.* 194, 276–282. <https://doi.org/10.1016/j.biortech.2015.06.112>.

Mertenat, A., Diener, S., Zurbrugg, C., 2019. Black soldier fly biowaste treatment - assessment of global warming potential. *Waste Manage.* 84, 173–181. <https://doi.org/10.1016/j.wasman.2018.11.040>.

Møller, J., Boldrin, A., Christensen, T.H., 2009. Anaerobic digestion and digestate use: accounting of greenhouse gases and global warming contribution. *Waste Manage. Res.* 27, 813–824. <https://doi.org/10.1177/0734242X09344876>.

Paredes, C., Bernal, M.P., Cegarra, J., Roig, A., Navarro, A.F., 1996. Nitrogen transformation during the composting of different organic wastes. In: Van Cleemput, O., Hofman, G., Vermoesen, A. (Eds.), *Progress in Nitrogen Cycling*

- Studies: Proceedings of the 8th Nitrogen Workshop held at the University of Ghent, 1994. Springer, Netherlands, Dordrecht, pp. 121–125. https://doi.org/10.1007/978-94-011-5450-5_19, 5–8 September.
- Perednia, D.A., 2017. A comparison of the greenhouse gas production of black soldier fly larvae versus aerobic microbial decomposition of an organic feed material. *Res. Rev. J. Ecol. Environ. Sci.* 5, 1–7.
- Pipatti, R., Wagner, J., Alves, S., Gao, Q., López Cabrera, C., Mareckova, K., Oonk, H., Scheehle, E., Sharma, C., Smith, A., Svardal, P., Yamada, M., 2006. Biological Treatment of Solid Waste, in: Eggleston H.S., B.L., Miwa K., Ngara T., Tanabe K. (Ed.), 2006 IPCC Guidelines for National Greenhouse Gas Inventories. IGES, Japan. <https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol5.html>.
- R Core Team, 2016. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>.
- Rynk, R., Richard, T.L., 2001. Commercial Compost Production Systems. In: Stoffella, P.J., Kahn, B.A. (Eds.), *Compost Utilization in Horticultural Cropping Systems*. Lewis Publishers, Boca Raton, Fla, pp. 51–93. <https://doi.org/10.1201/9781420026221>.
- Salomone, R., Saija, G., Mondello, G., Giannetto, A., Fasulo, S., Savastano, D., 2017. Environmental impact of food waste bioconversion by insects: application of life cycle assessment to process using hermetia illucens. *J. Clean. Prod.* 140, 890–905. <https://doi.org/10.1016/j.jclepro.2016.06.154>.
- Thanh Phong, N., Clemens, J., Cuhls, C., 2016. Greenhouse gas emissions from anaerobic digestion plants. *Vietnam J. Sci. Technol.* 54, 208–215. <https://doi.org/10.15625/2525-2518/54/4B/12043>.
- Varotto Boccazzi, I., Ottoboni, M., Martin, E., Comandatore, F., Vallone, L., Sprangers, T., Eeckhout, M., Mereghetti, V., Pinotti, L., Epis, S., 2017. A survey of the mycobiota associated with larvae of the black soldier fly (*Hermetia illucens*) reared for feed production. *PLoS One* 12, e0182533. <https://doi.org/10.1371/journal.pone.0182533>.
- Xiao, X., Mazza, L., Yu, Y., Cai, M., Zheng, L., Tomberlin, J.K., Yu, J., van Huis, A., Yu, Z., Fasulo, S., Zhang, J., 2018. Efficient co-conversion process of chicken manure into protein feed and organic fertilizer by *Hermetia illucens* L. (Diptera: Stratiomyidae) larvae and functional bacteria. *J. Environ. Manage.* 217, 668–676. <https://doi.org/10.1016/j.jenvman.2018.03.122>.
- Yu, G.H., Cheng, P., Chen, Y.H., Li, Y.J., Yang, Z.H., Chen, Y.F., Tomberlin, J.K., 2011. Inoculating poultry manure with companion bacteria influences growth and development of black soldier fly (Diptera: Stratiomyidae) larvae. *Environ. Entomol.* 40, 30–35. <https://doi.org/10.1603/En10126>.
- Zheng, L., Crippen, T.L., Holmes, L., Singh, B., Pimsler, M.L., Benbow, M.E., Tarone, A. M., Dowd, S., Yu, Z., Vanlaerhoven, S.L., Wood, T.K., Tomberlin, J.K., 2013. Bacteria mediate oviposition by the black soldier fly, *Hermetia illucens* (L.) (Diptera: Stratiomyidae). *Sci. Rep.* 3, 2563. <https://doi.org/10.1038/srep02563>.